

Museum Automation System

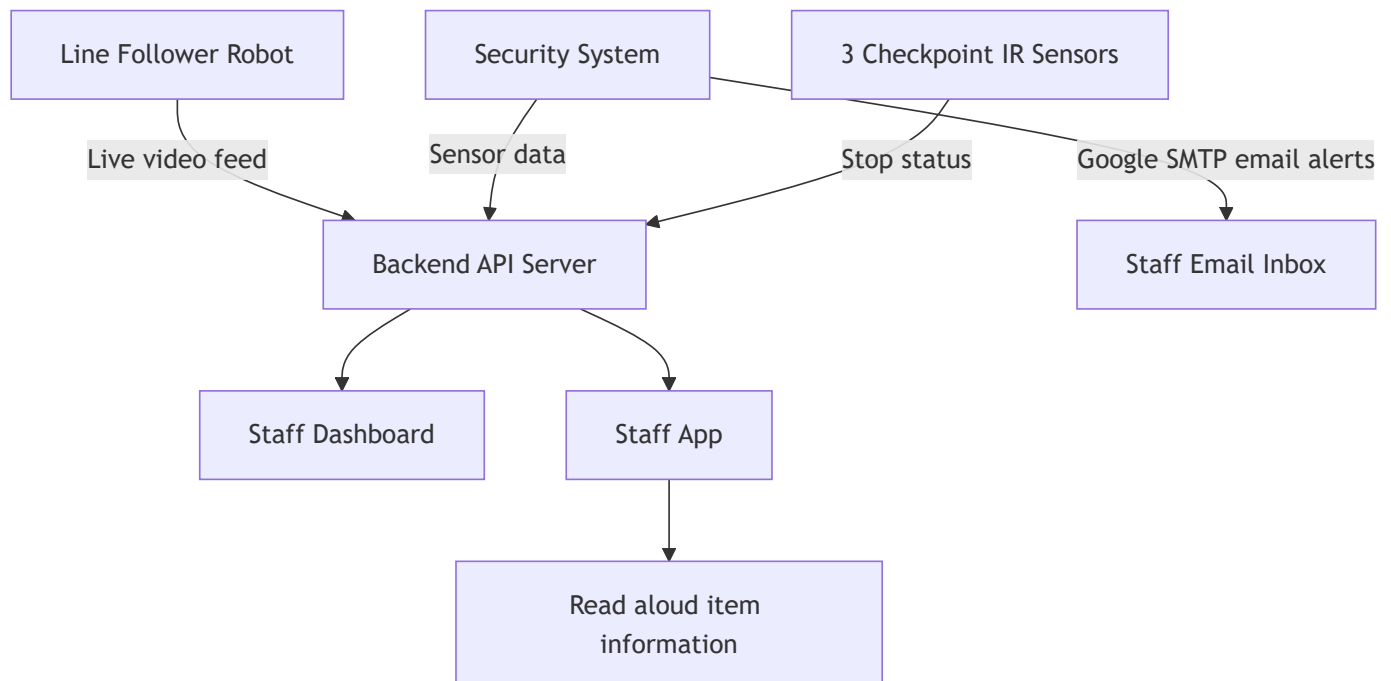
Simple system overview for your museum robot and security setup.

System Goal

The system helps visitors learn about historical items while improving museum safety.

- A robot follows a floor line.
- It stops at checkpoints.
- The camera scans a QR code at each stop.
- The server gets the text link from the QR code.
- The staff app reads the item information aloud.
- A security system watches motion, smoke, temperature, and humidity.

High Level Architecture



Robot Hardware

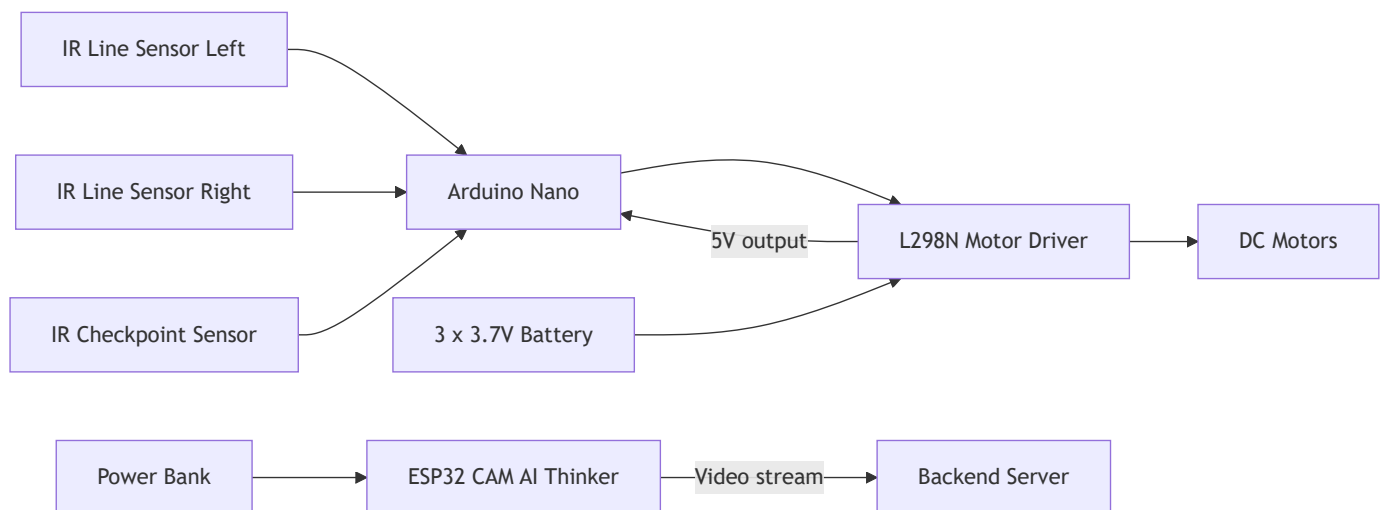
Main Controllers

- Arduino Nano
 - 2 IR sensors for line following
 - 1 IR sensor for checkpoint detection
 - Connected to L298N motor driver
- ESP32 CAM AI Thinker
 - Live video streaming to server

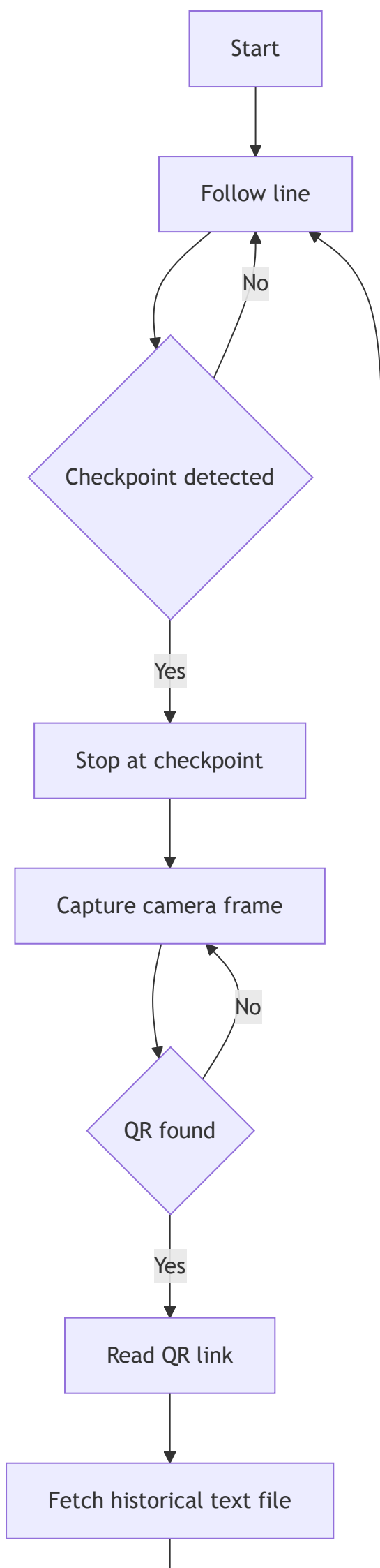
Power Design

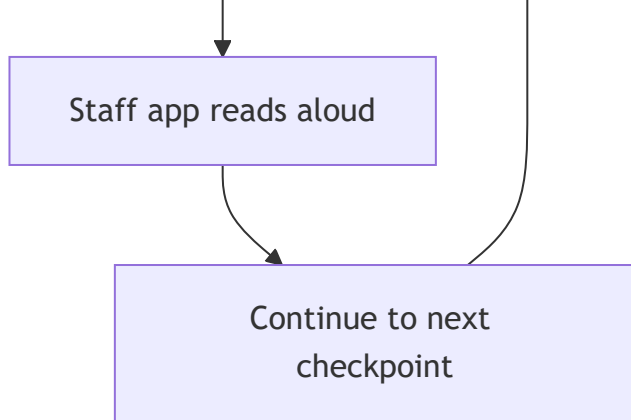
- Power bank powers ESP32 CAM board.
- 3 x 3.7V battery pack powers L298N and motors.
- Arduino Nano gets 5V from the L298N driver board.

Robot Wiring Diagram



Robot Behavior

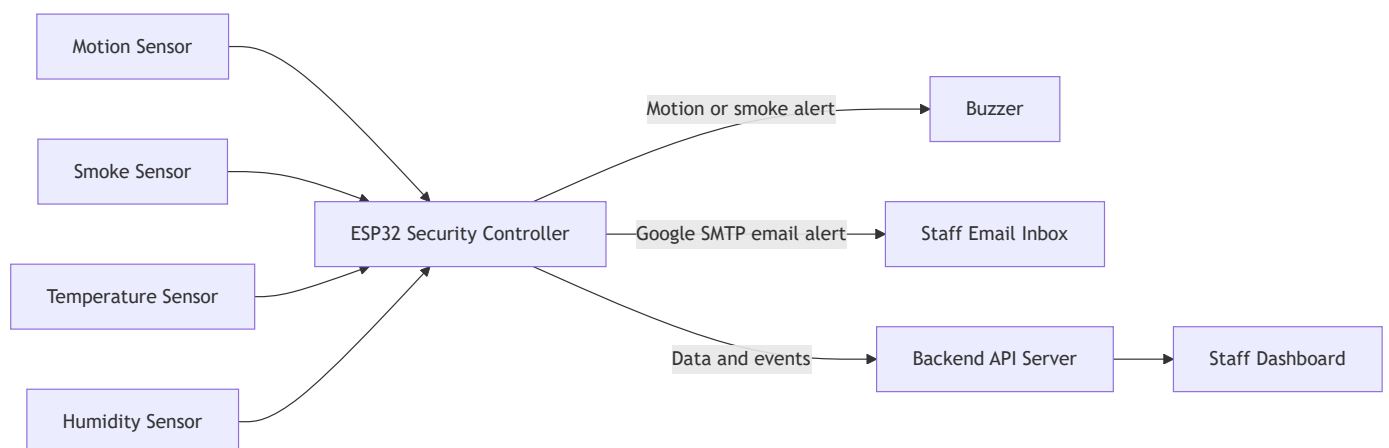




Security System

The security system reports to the same server.

- Detects motion
- Detects smoke
- Reports temperature
- Reports humidity
- Plays buzzer when motion or smoke is detected
- ESP32 security board sends email alert with Google SMTP when motion or smoke is detected
- Uses 3 checkpoint IR sensors to report robot stop status

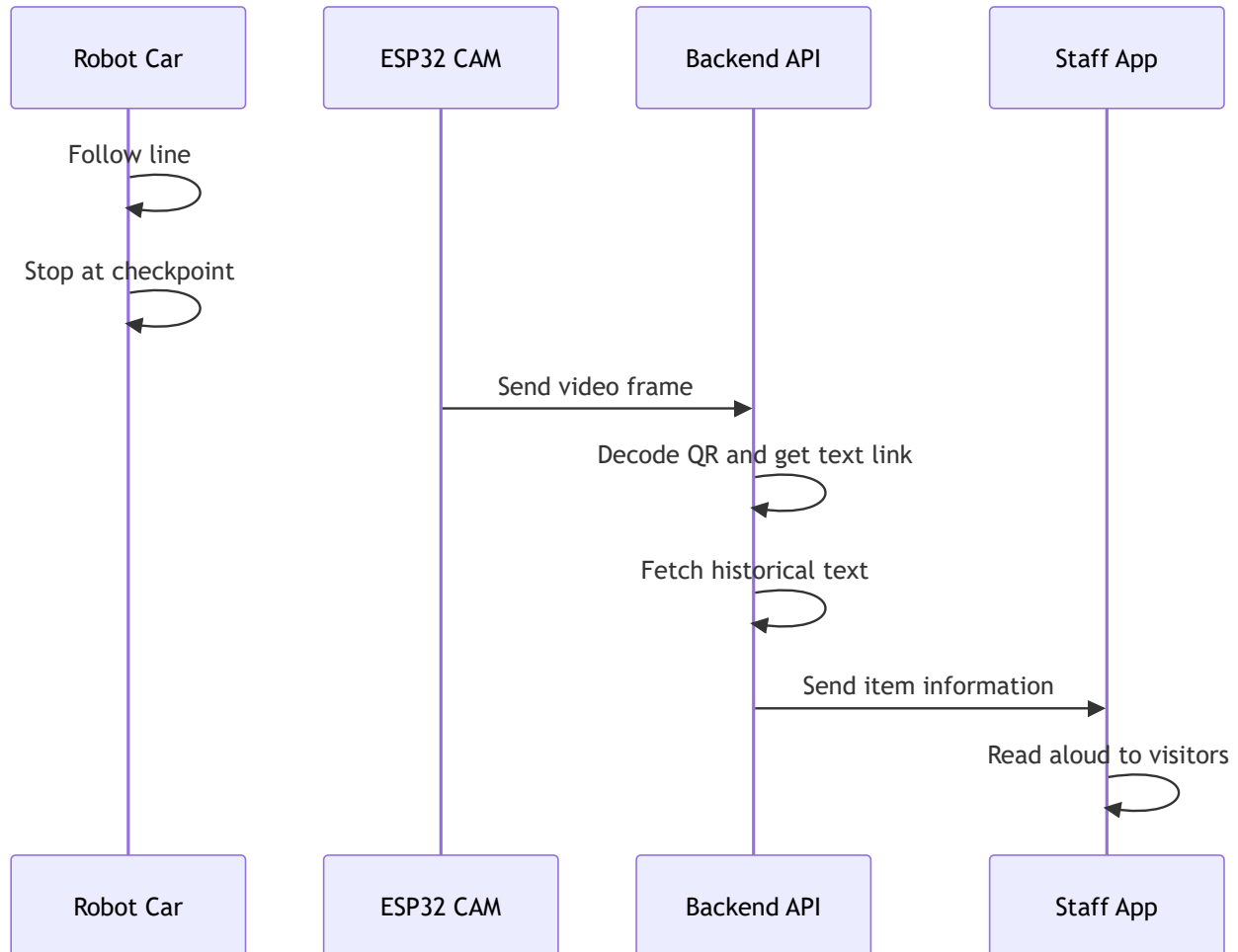


Server Features

- Backend API for robot and security data
- Dashboard for staff to monitor alerts and environment
- Video feed capture from ESP32 CAM
- Security alert events from ESP32 motion and smoke detection
- Google SMTP email alerts to staff for motion and smoke

- QR decode when robot is stopped
- Read text link from QR and fetch historical item text
- Send content to staff app for read aloud

Simple End to End Flow



Notes

- This design keeps robot control and video processing separated.
- It supports guided tours and live security monitoring at the same time.